

CSC4026Z Chat Server Documentation

Session Messages

These messages manage your connection to the server. The definitions below use [these type indicators](#).

Remember that your Client sends the Requests, the Server sends the Responses.

A trivial example of sending a *PING* message is included on the [Architecture page](#)

CONNECT

This is the first message you **MUST** send. It **MUST NOT** include a `session`. When the server responds, you **MUST** store the `session` value it returns, and include that value in every subsequent request your client makes. The server also assigns you a username. You **MAY** change it with [SET_USERNAME](#).

Request

```
CONNECT_request {
    u8 request_type = 1
    u32 request_handle      # Always include a random request_handle
}
```

Response

```
CONNECT_response {
    u8    response_type = 22
    u32    session      # The Server fills in a random value, you
    u32    response_handle
    s[500] message      # A Welcome message from the Server
    s[20]  username     # Server-generate username for you
}
```

PING

Use these messages to maintain an ongoing session with the server. The server will mark you as disconnected and forget your connection if it hasn't heard from you in 60 seconds. Send a *PING* message every 30 seconds to ensure that the server hears from you (and in case one of your *PING*'s was lost.)

Request

```
PING_request {  
    u8  request_type = 3  
    u32 session  
    u32 request_handle  
}
```

Response

```
PING_response {  
    u8  response_type = 24  
    u32 session  
    u32 response_handle  
}
```

DISCONNECT

You **MAY** send a disconnect when your client exits. If you don't send a disconnect, then your server session will timeout within 60 seconds.

Request

```
DISCONNECT_request {  
    u8  request_type = 2  
    u32 session  
    u32 request_handle  
}
```

Response

```
DISCONNECT_response {  
    u8      response_type = 23  
    u32     session  
    u32     response_handle  
    s[200] message          # A farewell message from the Server  
}
```

OK

This is a simple message to indicate the server has seen and processed some request.

Response

```
OK_response {  
    u8      response_type = 21  
    u32     session  
    u32     response_handle  
}
```

ERROR

Errors are returned by the Server if the Client sends invalid data. Clients **MUST NOT** send Error messages to the Server.

Response

```
ERROR_response {  
    u8      response_type = 20  
    u32      session  
    u32      response_handle  
    s[1000] error          # An error message from the Server  
}
```

SERVER_MESSAGE

Server Messages may be sent by the server from time to time. These are not associated with any user, but can be used by the Server to broadcast messages from the Admin. All connected users receive the same *SERVER_MESSAGE* when one is sent.

To test this out, the server sends lines from Shakespeare's sonnets every 5 minutes to all connected users.

Response

```
SERVER_MESSAGE_response {  
    u8      response_type = 36  
    u32      session  
    u32      response_handle = None  
    s[500] message          # A broadcast message from the Server  
}
```

SERVER_SHUTDOWN

This message type indicates that the server is shutting down, and this is the last message you will receive from the Server. When you receive this message, you **SHOULD** destroy any session state your client currently holds, as the server does not maintain state between invocations. You **MAY** attempt to reconnect to the Server within a short time period, although there is no guarantee the Server has restarted.

Response

```
SERVER_SHUTDOWN_response {  
    u8      response_type = 37  
    u32      session  
    u32      response_handle = None  
    s[500] message          # A shutdown message from the Server  
}
```

```
}  
}
```

Move on to the [Channel messages](#).

Documentation built with [MkDocs](#).